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DNA methylation-based forensic tissue identification

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ABSTRACT

Identifying the source tissue of biological material found at crime scenes can be very informative in a number of cases. Despite their usefulness, current visual, catalytic, enzymatic, and immunologic tests for presumptive and confirmatory tissue identification are applicable only to a subset of samples, might suffer limitations such as low specificity, lack of sensitivity, and are substantially impacted by environmental insults. Moreover these assays are incompatible and thus cannot be multiplexed. Thus they are less amenable to automation. In addition their results are operator-dependent. A better alternative approach is tissue identification based on messenger RNA (mRNA) or microRNA (miRNA); however, RNA is not as stable as DNA, and requires the use of non-standard procedures by forensic laboratories. Herein a DNA-based assay is described that enables tissue identification based on detection of tissue-specific methylation patterns. DNA samples are subjected to digestion by a methylation-sensitive restriction endonuclease followed by multiplex amplification of specific genomic targets with fluorescent-labeled primers, capillary electrophoresis of amplification products, and automatic signal analysis by dedicated software, yielding the source tissue of the sample. The single tube assay was designed for easy integration by forensic laboratories (as the assay utilizes the same platforms as current forensic STR profiling). The system is fully automatable, provides operator-independent results, and allows combining tissue identification with profiling in a single procedure. The assay was tested on 50 DNA samples from blood, saliva, semen, and skin epidermis, and the source tissue was successfully identified in all cases. Detection of semen and DNA profiling were combined into one assay and the ability to detect mixtures of semen and saliva in various ratios was demonstrated. The assay correctly detected semen in all samples where it was present, and the calculated percentage of semen was comparable to the fraction of semen in the samples. The results demonstrate that methylation-based tissue identification is more than a proof-of-concept. The methodology holds promise as another viable forensic DNA analysis tool for characterization of biological materials.

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1. Introduction

Identification of the source tissue of biological material found at crime scenes can be very informative in a number of forensic cases [1–4]. For more than a century [1] presumptive and confirmatory tests have been sought and developed for body fluid identification [5]. Presumptive tests are used as screening tests but tend to have specificity limitations. Confirmatory tests are used for absolute tissue identification and can be useful in reconstructing the events of a crime. Both sets of tests are based on protein-based enzymatic or immunologic assays (as well as visualization for sperm identification), and thus depend on stability of the target molecules for detection of the activity or presence of a protein that is specific for the tissues of interest. Examples of more recently developed protein-based commercial kits include Bluestar® for detection of hemoglobin, Phadebas® for detection of amylase, and ABAcard® p30 for detection of the p30 antigen. Protein-based assays have some advantages over many of the classic methods and have successfully replaced many of them, although some techniques such as microscopic identification of sperm cells, which is still considered the most reliable and widely accepted confirmatory technique for detection of semen, remain widely used [5]. Despite their utility in forensic analyses, protein based methods suffer from several limitations. Proteins tend to be less stable than DNA; thus DNA may persist longer than the proteins and a negative result may not correlate with the amount of DNA in the sample. These assays rely on the presence of intact and/or functional protein and might not work on degraded or environmentally damaged samples [6]. Low specificity is another limitation associated with many presumptive and a number of confirmatory tests, due to cross reaction with other molecular species or tissues [7]. Most commercial kits provide non-quantitative outputs, typically indicating presence or absence of the tested material in the sample without a well defined threshold;

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therefore the tests can suffer from subjective, user-dependent and/or ambiguous results and an inability to quantify the contributions from multiple tissue sources. Another consequence of a non-quantitative assay is that there is no statistical confidence associated with either a positive or negative result. More important though is the fact that most protein-based assays consume sample, which in turn reduces the available amount of precious DNA evidence, a concern that is especially relevant to the analysis of already challenged limited quantity samples. Lastly, most tests, including the commercially available kits employ different types of non-standardized formats making it difficult to multiplex and automate procedures. Thus, the protein-based systems require manual labor, carrying with them the risk of increased human error.

An alternative approach to protein-based assays is analysis of tissue-specific species of mRNA [2,3,6–10] and recently reported, miRNA [11]. Because of differential expression of genes in different tissues, levels of mRNA and miRNA species can be indicative of tissue source and represent better target molecules that can potentially overcome some of the limitations associated with protein-based assays. However, RNA-based assays still suffer from the fact that RNA is less stable that DNA because of the ubiquitously present RNases [7] and require additional consumption of samples, unless the RNA is co-extracted with DNA.

Two recent approaches that show promise are fluorescence [12], and Raman [13] spectroscopy. They offer the potential of on-field, non-destructive approaches for forensic tissue identification. However, tissue specificity may still be problematic due to cross reaction with biological material from other species [14] and with non-biological material that may be present in the samples.

A more desirable tissue identification assay is one that does not necessarily have to consume additional sample, can be more directly related to the amount and persistence of DNA, and has similar limits of detection.A DNA-methylation based assay for forensic tissue identification can potentially overcome the limitations associated with extant methods. In mammalian DNA, methylation occurs at the C5 position of cytosine in some CpG dinucleotides [15]. In the human genome 70–80% of all CpGs are methylated, while unmethylated CpGs are grouped in "CpG islands" positioned at the 5′ ends of many human genes [16]. DNA methylation is generally believed to inhibit gene expression, probably by affecting chromatin structure [17], although in some cases it is associated with gene activation [18]. Regardless of the mechanism by which methylation affects gene expression, which remains poorly understood [19], differential methylation patterns between different tissues have been demonstrated [18,19]. Frumkin et al. found genomic loci that are consistently methylated and other loci that are consistently unmethylated in natural human DNA samples extracted from forensically relevant tissues [20]. They showed that methylation patterns could be used to differentiate between natural and artificially synthesized DNA. Herein we utilize genomic loci that we found to be differentially methylated between forensically relevant tissues that could form the foundation of a DNA-based forensic tissue identification assay. The approach potentially offers tissue specificity comparable to RNA-based assays, similar sensitivity of detection to that of DNA profiling, and no additional consumption of sample as a methylation-based assay can be multiplexed with existing STR typing protocols.

2. Materials and methods

2.1. Collection of biological tissues

Blood (venous/menstrual), saliva, semen, skin epidermis, urine, and vaginal secretion were collected from volunteers. Informed consent was obtained from all participants recruited into the study.

2.2. DNA extraction and quantification

DNA was extracted from all samples using the organic extraction protocol of Fritsch et al. [21]. The quantity of recovered DNA was determined using the Quantifiler® Human DNA quantification kit (Applied Biosystems) and the 7300 Real-Time PCR system (Applied Biosystems).

2.3. Selection of genetic loci and primer design

Random CpG islands (defined as a region $\geq$200 bp and with a GC content $\geq$0.5 and with at least 8 CGs; corresponding to observed/expected CpG ratio $>$ 0.6) were searched using a software program (implemented in MATLAB, MathWorks) developed for this task. From these, CpG islands that contained a HhaI recognition sequence were selected for initial screening. Primers (18–28 bps) flanking the HhaI recognition sequence were designed with a Tm of 64–66 °C and for amplicon sizes of 66–150 bps. Fluorescently labeled forward primers (FAM or JOE) were purchased from Integrated DNA Technologies. Loci were screened in pairs for differential methylation by comparing signals obtained from amplification of pooled DNA samples (each containing DNA from 10 individuals) of each tissue type. Loci that showed significant differential amplification patterns (defined as the largest methylation ratio observed in all tissue types being greater that three times the smallest ratio observed) were selected for multiplexing. In total, 205 genomic loci were screened of which 38 showed significant differential amplification patterns. From these, 16 loci were used in the experiments. For the stand-alone tissue identification assay, 15 loci that could be co-amplified without significant noise were used. For the combined semen detection and profiling assay two loci (L68346 which was also used for the stand-alone assay and L16264) were used.

2.4. Endonuclease digestion, PCR, and capillary electrophoresis

In stand-alone assay experiments, each DNA sample was digested by HhaI and subjected to PCR amplification in the same reaction, consisting of 30 U HhaI (New England Biolabs), 2.5 U AmpliTaq Gold (Applied Biosystems), 2.5 μg BSA (New England Biolabs), 0.2 mM each dNTP, 0.1–0.3 μM each primer (see Tables 1 and 4 for primer concentrations), 2.5 μl reaction buffer (150 mM Tris–HCl, 15 mM $MgCl_2$) and DDW to a total volume of 25 μl. The reaction composition in the combined tissue identification-profiling experiments was the same as in the stand-alone assay except for the primers, which were 0.2 μM each semen identification primer (Table 4) together with 5 μl Primer Set of ProfilerPlus (Applied Biosystems). Reactions (for both stand-alone and combined assay) were performed in a GeneAmp® PCR System 9700 (Applied Biosystems), and the thermocycling program used was: 37 °C for 15 min, 95 °C for 11 min, followed by 28 cycles of 94 °C for 1 min, 59 °C for 1 min, 72 °C for 1 min, and followed by a final extension step of 60 °C for 45 min. A mix containing 1.5 μl of amplification products, 24.5 μl HiDi formamide (Applied Biosystems) and 0.5 μl ROX size standard (Applied Biosystems) was denatured (95 °C for 3 min immediately followed by 3 min on ice), and run on an ABI 310 Genetic Analyzer (Applied Biosystems) according to the manufacturer's instructions. All resulting electropherograms were analyzed using in-house developed software, and electropherograms resulting from combined tissue identification-profiling experiments were also analyzed using GeneMapper ID-X analysis software (Applied Biosystems).

2.5. Tissue identification assay—data analysis

Data analysis for tissue identification (including in the combined semen detection and profiling) was performed by a

Table 1
Panel of loci used for tissue identification assay.

#	Locus	Location[a]	Forward primer	Reverse primer	Size[b]	Conc.[c]
1	L91762	Chr12 (73985697)	GCAGCAGGCCGCGGAGAAG (FAM)	AGCAGCTGTGCCGGGCCAG	66	0.2 μM
2	L68346	Chr3 (12175317)	CAGCAACAGCACCCAGCTTG (JOE)	CACAGGCTCAGTCGCGGATC	70	0.2 μM
3	L50468	Chr3 (55492965)	AGGAAACCTCAGTAGCAAAATTG (JOE)	GCGAGACTTTAGGTGTGCATC	75	0.2 μM
4	L14432	Chr22 (29866149)	CGTAGGCTGCGGTGAGCTC (FAM)	GATCCATGCCCGCTGGGATG	75	0.2 μM
5	L4648	Chr1 (35815331)	CAGCCTAGACGTCAAGTTACAG (JOE)	ACGACCTCCGGATCCAACTG	80	0.2 μM
6	L39664	Chr17 (53710330)	CCCAGCTGGTTGGACATGTTG (FAM)	CACTTCCTTCGTGGACGCC	82	0.2 μM
7	L30139	Chr17 (36996489)	GAGAAGCGGGAGGATGAGAC (FAM)	CCGCATCTCCTCCGTCCTG	84	0.2 μM
8	L55429	Chr5 (1548043)	GCCTTCAGCAGGAAGTCCAC (JOE)	CCTGTGCCTCACACAGACTC	88	0.2 μM
9	L62086	Chr19 (40478538)	GTGCATGGTGTCTGGTACTTC (FAM)	GAAGCTCTCGTGGACTACTTG	89	0.1 μM
10	L76138	Chr19 (3130217)	CAGCCTGCTCTTCACTGCAG (JOE)	AGAGGCCGATGAAGCCGTAG	100	0.2 μM
11	L15952	Chr7 (2741305)	CTCCCTGATTTACGACAAGTTC (FAM)	GACAGTATGCTGATGCTTCTTG	117	0.2 μM
12	L36599	Chr19 (4867642)	AAGGGCAGAGTTCCGCTGTC (FAM)	CGGATGCAGGAGGATCCTAG	130	0.2 μM
13	L26688	Chr17 (77844826)	CGGACCAGATTGCTGGTCAC (JOE)	CGACCTTGCCAGATGTTTGAC	134	0.2 μM
14	L81528	Chr19 (47395603)	AGCCTCATCCACACTGACCAG (JOE)	TCAGAGCTCTCCTATCTGGAC	141	0.2 μM
15	L36556	Chr19 (50962118)	GCCAGGCCGTTGATGATGAC (JOE)	GAATATGGAGCCCTGGGCAG	149	0.3 μM

[a] Genomic location of the examined CG according to UCSC version hg18 is provided in brackets following the chromosome number.
[b] In bps.
[c] The concentration refers to both the forward and reverse primers.

MATLAB (MathWorks)-based in-house developed software that accepts as input fsa files and outputs the most likely source tissue. The software's algorithm is provided in supplementary material (Text S1).

Derivation of semen percentage in mixture samples was performed based on profiling loci and tissue identification loci. Derivation based on profiling loci was performed by dividing the sum of rfu values of alleles corresponding to the contributor of the semen by the total sum of rfu values of all alleles in STR loci that had 4 genotyped alleles. Derivation based on tissue identification loci was performed first by calculating the expected methylation ratio for all possible percentages of mixtures (from 0 to 100% with 1% increments, based on the methylation ratio of undigested DNA), and the algorithm provided the percentage of semen for which the expected methylation ratio is closest to the observed methylation ratio.

2.6. Simulation of degraded samples

Twenty nanograms of each sample were digested for 10 min at 37 °C with 0.01 U DNAseI (Ambion), 2 μl 10× DNAseI buffer (Ambion) in total reaction volume of 20 μl, followed by heat-inactivation at 75 °C for 10 min. One nanogram of each digested DNA sample was then subjected to the combined tissue identification-profiling assay.

3. Results

3.1. Description of tissue identification assay

The tissue identification assay uses a panel of loci that are differentially methylated between tissues to determine the most probable source tissue of a DNA sample (Table 1). A scheme of the assay is presented in Fig. 1A. One nanogram of DNA from a forensic sample in question is digested with the HhaI methylation-sensitive restriction enzyme, which cleaves DNA at its recognition sequence GCGC only if it is unmethylated (while leaving methylated targets intact). A panel of tissue identification loci is then amplified by PCR from the digested DNA using fluorescently labeled primers, and an aliquot of amplified products is separated by capillary electrophoresis. Loci with higher methylation levels are amplified with higher efficiency (because more DNA molecules are protected from digestion), producing a relatively strong signal in the electropherogram (Fig. 1B, locus A). Conversely, loci with a lower methylation level are amplified with lower efficiency, yielding a relatively weak signal in the electropherogram (Fig. 1B, locus B).

Automated signal analysis software enables analysis of an output electropherogram, and assigns heights (in rfu) to amplicons corresponding to tissue identification loci. Although the height of a single locus is correlated with its methylation level, the actual methylation level cannot be derived, because that level also depends on the precise template concentration and specific PCR conditions. Instead, ratios of methylation levels between co-amplified loci can be calculated, since all the loci are subjected to the same template concentration and specific reaction conditions. Therefore, for each pair of loci a methylation ratio, defined as the ratio between the heights of the first and second loci, is calculated, and this ratio reflects the ratio between the methylation levels of the corresponding loci. All calculated methylation ratios are combined into a single vector (if, for example, in a 15-loci panel the vector contains 105 ratios—between locus 1 and 2, locus 1 and 3, etc.), which is then compared to a database of reference vectors obtained from a data set of samples of known tissue origin. The tissue identification algorithm calculates for each potential tissue source a likelihood score, reflecting the likelihood that the DNA sample originated from that tissue, and the output of the algorithm is the most likely tissue.

3.2. Analysis of noise factors affecting methylation ratios

The assay described herein is based on utilizing loci that are differentially methylated between tissues, for which methylation ratio values are generally tissue-specific (Fig. 1C). However, even in differentially methylated tissues, there potentially could be some overlap in the observed methylation ratios as a result of natural variability in methylation levels among different individuals and/or artifacts associated with the PCR such as differences in DNA template concentration (e.g. as a result of pipetting errors and stochastic effects). The effect of each of these "noise" factors was analyzed in relation to the "signal", which is the average methylation ratio difference between tissues. Two loci with an average 10-fold difference in methylation ratio values between semen and blood were selected. Different PCRs were performed on the same semen sample using 1 ng and 2 ng DNA as template. The ratios obtained from these amplifications were compared to each other and to ratios obtained from amplification of different blood samples. The differences between the ratios obtained in different PCRs and using different amounts of input DNA were more than one order of magnitude smaller than the differences between the ratios obtained from the different tissues (Fig. 1D).

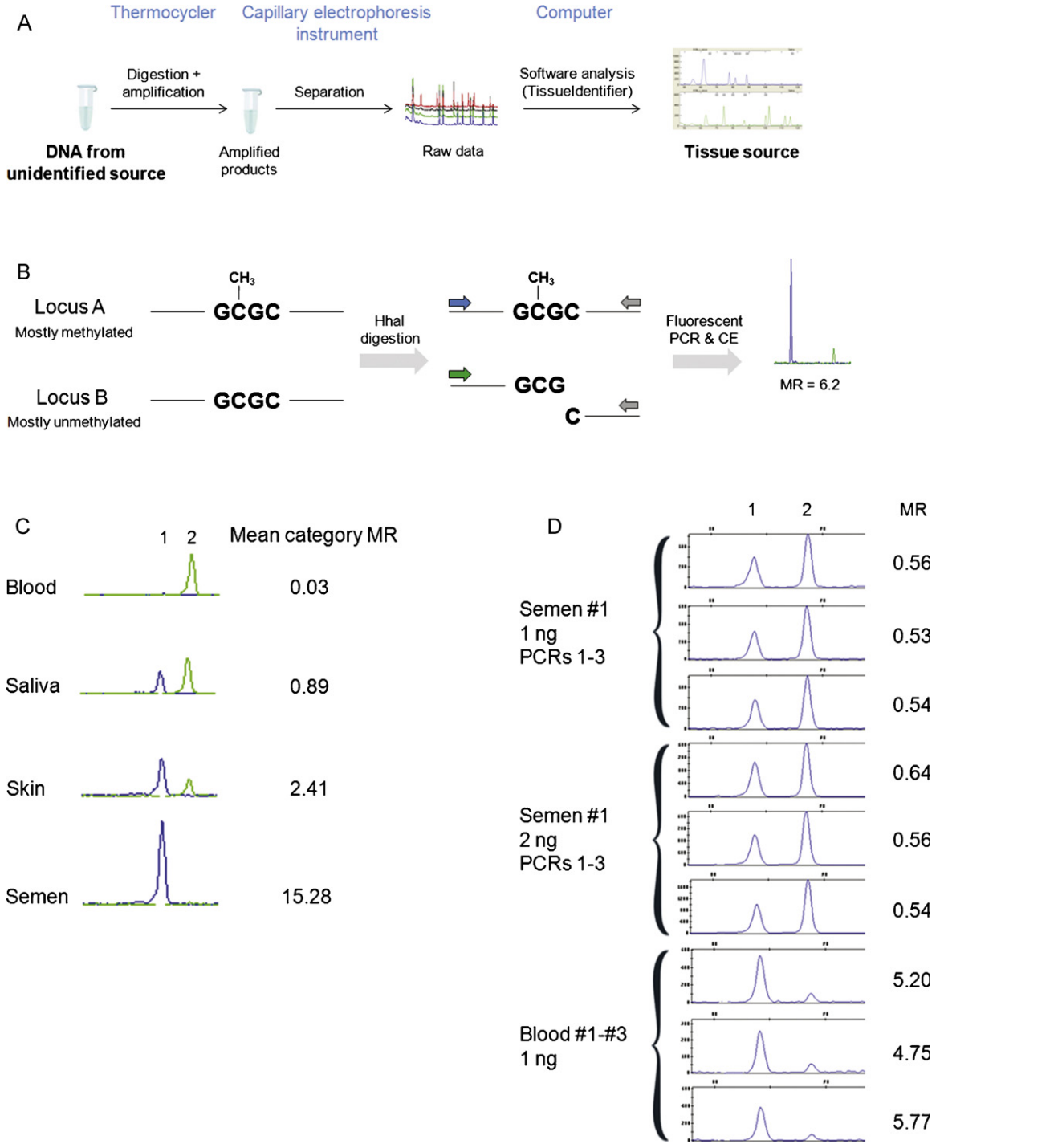


Fig. 1. Tissue identification assay. (A) Schematic overview of the assay. (B) Biochemical procedure—methylated loci remain intact during digestion and subsequently are amplified efficiently in the PCR, producing a strong signal (locus A) while unmethylated loci are digested and subsequently amplify inefficiently in the PCR, producing a weak signal (locus B). The methylation ratio (MR; rfu of locus A/rfu of locus B) reflects the differential methylation level between loci A and B. (C) Methylation ratios between locus 1 and locus 2 are different in blood, saliva, skin, and semen, reflecting the differential methylation patterns in these tissues. (D) The observed differences in methylation ratios between blood and semen (i.e. the signal) are more than a magnitude greater than the observed differences in methylation ratios obtained from different PCRs and from different amounts of input DNA (i.e. the noise).

3.3. Stand-alone tissue identification assay

A stand-alone tissue identification assay was performed on 50 samples: 14 blood, 14 saliva, 11 semen, and 11 skin epidermis using a panel of 15 tissue identification loci, ranging in amplicon size from 66 to 150 bps. Fig. 2 depicts electropherograms of eight of these samples—two of each tissue type. Although samples of the same tissue type from different individuals were not completely identical (due to natural variability of methylation levels and stochastic PCR effects), it was nevertheless evident that each tissue type had a distinct methylation profile. For example, the ratio of L91762/L68346 in semen samples ranged from 0.04 to 0.53 and was higher in all other tissue samples (2.15–18.28; Table 2 and supplementary material Fig. S3), and therefore a low L91762/L68346 ratio was distinctive of semen samples. The ratio of L76138/L26688 was low in blood and saliva (0.08–1.54) and higher

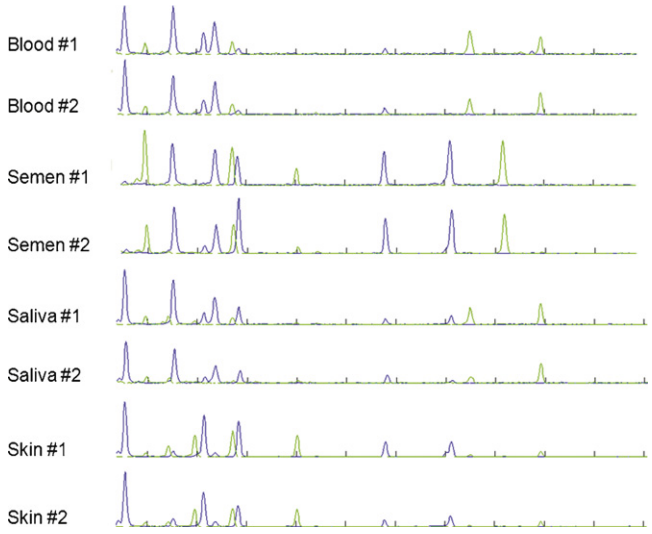


Fig. 2. Tissue identification of different tissue types. Electropherograms of 8 samples (2 samples from each of 4 different tissues) are depicted demonstrating the distinct amplification pattern observed for each tissue type.

Table 2
Methylation ratios of L91762/L68346 and L76138/L26688.

	L91762/L68346		L76138/L26688	
	Mean ± std	Range	Mean ± std	Range
Semen	0.16 ± 0.14	0.04–0.53	6.50 ± 5.80	2.04–19.00
Blood	4.36 ± 1.69	2.15–7.73	0.30 ± 0.20	0.08–0.76
Saliva	7.10 ± 3.30	3.36–15.38	0.56 ± 0.41	0.16–1.54
Skin epidermis	11.40 ± 4.20	4.60–18.28	8.41 ± 5.58	2.1–19.56

in semen and skin epidermis (2.04–19.56; Table 2 and supplementary material Fig. S4); therefore a high ratio of L91762/L68346 concomitant with a high ratio of L76138/L26688 was distinctive of skin epidermis samples. The tissue identification algorithm correctly identified the true tissue source of all 50 samples (supplementary material Table S5). Further analysis of the data using subsets of tissue identification loci revealed that the full set of 15 loci was redundant and 100% identification was achieved by using only 7 out of the 15 loci (L91762, L68346, L50468, L14432, L30139, L15952, and L26688). Because tissue identification relies on efficient digestion, complete digestion of template DNA by HhaI was assessed in a single digestion-amplification reaction setup. The 50 DNA samples were subjected to the same procedure but using a panel of two different loci (Table 3)—a control locus without a HhaI site (L98328) and a locus which was previously shown to be completely unmethylated in these tissues (SW14; Ref. [20]). All samples showed presence of the reference locus and complete absence of SW14, indicating complete digestion by HhaI in contrast to the undigested sample which showed presence of both loci (example runs are shown in supplementary material Fig. S2).

3.4. Combined semen detection and profiling

The tissue identification procedure utilizes the same platforms used in standard STR profiling (i.e. thermocycler and capillary electrophoresis instruments) and is therefore readily amenable to integration with DNA profiling in a single reaction. This capability was demonstrated by generating an integrated STR profiling and semen detection assay. A STR profiling kit (Profiler Plus) PCR was supplemented with primers specific for two semen identification loci (Table 4). These loci consist of a 70 bps amplicon which is efficiently amplified only in semen and a 95 bps amplicon which is efficiently amplified in all tissues except semen. The assay was tested on pure samples of semen, urine (male), venous blood, menstrual blood, vaginal secretion, and saliva. The correct pattern of semen identification amplification was observed and the tissue identification algorithm correctly identified presence/absence of semen in all samples. Fig. 3A and B depicts complete electropherograms of semen and urine samples (obtained from the same individual), and Fig. 3C depicts blue channel data from all samples including the two semen identification loci and three profiling loci. The observed methylation ratio was 25.04 in the semen sample, and 0.04–0.089 in the other samples. The combined assay was also tested on mixtures of semen and saliva at various ratios (Fig. 3D). For each of these mixtures the percentage of the semen in the sample was derived from the methylation ratio of the semen identification loci and compared with the corresponding percentage derived from the profiling loci (the semen and saliva samples were obtained from different individuals and could therefore be differentiated based on STR loci). The percentages derived were comparable, with a maximum difference of 10% (Table 5). Moreover, the presence vs. absence of semen was correctly determined in all samples (including a sample that contained only 13% semen; Fig. 3D). The STR profiles obtained from the samples using the integrated profiling-semen detection assay were identical to profiles obtained from the same samples using ProfilerPlus (without semen detection).

Table 3
Loci used for assessing HhaI digestion.

Locus	Location[a]	Forward primer	Reverse primer	Size (bp)
SW14	Chr7 (141084738)	GCGAAGGAAGTGCTGGAGTC (FAM)	GTTTCTTGAAAGGGCCAGACAC	94
L98328	Chr11 (86241803)	CAAAGTACTGGGGTTACAGGTG (FAM)	GGATGAACCTTTAAGACATCATC	97

[a] Genomic location of the examined CG according to UCSC version hg18 is provided in brackets following the chromosome number (in locus L98328, which does not have a HhaI recognition site, the location is of the beginning of the forward primer).

Table 4
Panel of loci used for semen detection.

Locus	Location[a]	Forward primer	Reverse primer	Size[b]	Conc.[c]
L68346	Chr3 (12175317)	CAGCAACAGCACCCAGCTTG (FAM)	CACAGGCTCAGTCGCGGATC	70	0.2 μM
L16264	Chr2 (2734005)	GGACGAGTTAACTTCCTTAATTTC (FAM)	GTTTCTTCGCGGAACCTGGTTTAACTTC	95	0.2 μM

[a] Genomic location of the examined CG according to UCSC version hg18 is provided in brackets following the chromosome number.
[b] In bps.
[c] The concentration refers to both the forward and reverse primers.

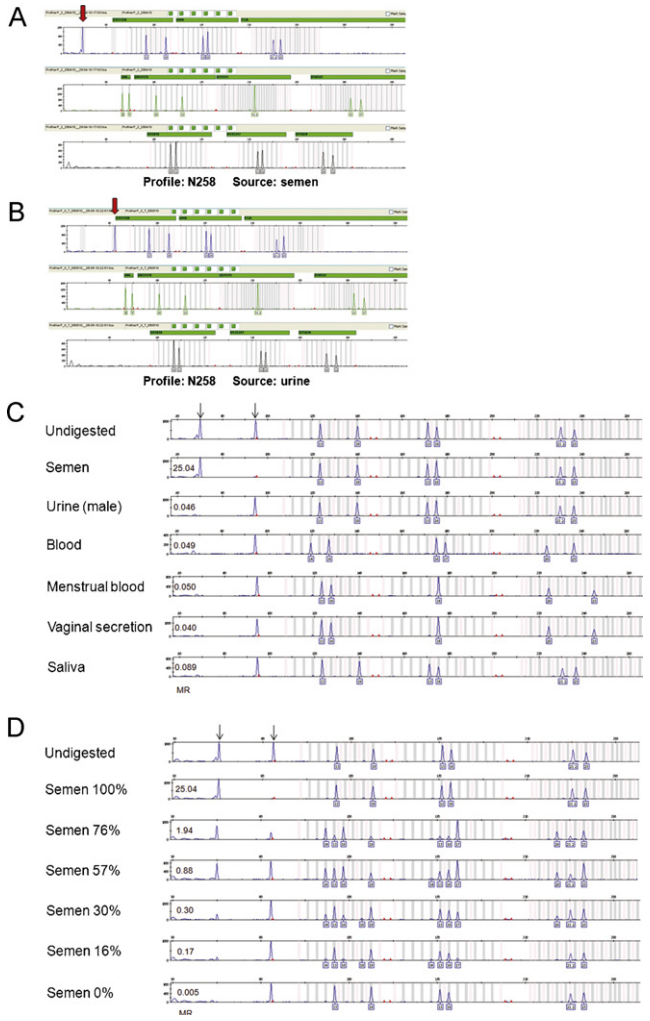


Fig. 3. Combined semen detection and DNA profiling. (A and B) Electropherograms of the combined assay performed on a semen (A) and urine (B) sample from the same individual. The ProfilerPlus profiles of the samples are identical, and the differences in tissue source are indicated by the differential amplification of two semen detection loci (red arrows). (C) Blue channel data of the combined assay performed on samples from different tissue types. The methylation ratio between the semen detection loci (arrows) is 25.04 in the semen sample, and less than 0.1 in all other tissue types. The top panel includes undigested DNA for reference. (D) Blue channel data of the combined assay performed on mixtures of semen and saliva (from different individuals) with various ratios. The methylation ratio is correlated to the percentage of semen in the mixture.

Table 5
Detection and quantification of semen component in semen-saliva mixtures.

Sample number	% semen in sample (according to STR typing)	% semen in sample (according to our assay)	Semen detected?
1	100	100	Yes
2	100	100	Yes
3	100	100	Yes
4	76	66	Yes
5	57	47	Yes
6	30	23	Yes
7	16	15	Yes
8	0	0	No
9	0	0	No
10	0	0	No

Each row represents a single reaction.

3.5. Aged and degraded samples

The stand-alone tissue identification assay was tested on two DNA samples extracted from 20-month-old stains (blood and semen on cotton) as a proof-of-concept for application to forensic-type samples. Both samples amplified well, similarly to fresh samples (supplementary material Fig. S6), and the algorithm correctly identified the tissue source of both samples. In order to simulate degraded DNA samples, two additional DNA samples (blood and semen) were subjected to digestion with DNAseI under conditions that yield partial digestion. These samples were then analyzed with the combined semen detection and profiling assay. DNAseI digestion resulted in dropout of alleles in the larger STRs in both samples such that their profiles were partial. In the blood sample only D3S1358 and Amelogenin alleles were typed, and in the semen sample dropout of alleles was observed in FGA, D21S11, D18S51, D13S317, and D7S820 (Supplementary material Fig. S7). The smaller tissue identification loci amplified successfully with the expected pattern, and the algorithm correctly identified the presence/absence of semen in these samples.

4. Discussion

The methylation-based assay described here unequivocally identified the tissue source in both single-source and mixture samples. The methodology has several features that make it a promising tool for forensic applications and thus might be worthwhile to pursue further for validation and implementation. Various aspects of the tissue identification assay are discussed below.

4.1. Potential tissue identification loci

In this study 205 HhaI recognition sequences in CpG islands were screened, and 38 of them were sufficiently differentially methylated between tissues to be useful for tissue identification. Depending on the precise definition used there are ~20,000 CpG islands in the human genome, and therefore there is potentially a large pool of CpG islands useful for tissue identification. In addition, in each island only one CpG site was examined; it is possible that islands that did not show differential methylation contain other CpG sites that are differentially methylated. Furthermore, CpG islands represent only a small fraction of the genome, and other genomic regions may also be informative for tissue identification. Thus, there are likely a large number of loci that could be useful for tissue source identification that could further enhance the assay described herein.

4.2. Informativeness of tissue identification loci

The informative nature of the tissue identification loci stems from their differential methylation between tissues. For example the methylation level of L91762 was much lower than the methylation level of L68346 in semen, whereas in blood, saliva, and skin epidermis it was higher. A low L91762/L68346 ratio indicated semen whereas a high L91762/L68346 ratio indicated other tissues. However, this ratio was also informative in the high range for differentiating between blood, saliva, and skin epidermis. In these tissue types there was some overlap in the observed distributions of ratios (supplementary material Fig. S3), but the distributions were not identical, with the mean ratio being lowest for blood, intermediate for saliva, and highest for skin epidermis. Therefore, for example, a ratio of 7, which falls within the observed ranges of all three tissue types, is more likely to indicate saliva than blood or skin epidermis, since it is close to the median of the distribution of saliva samples and is at the extreme right and left of

the distributions of blood and skin epidermis samples, respectively. Although the amount of information in such a single locus is limited, the tissue identification algorithm amalgamates such probabilistic information from the entire set of ratios and determines the most likely tissue with high probability. The situation is similar to DNA profiling, in which a single locus often does not carry substantial information, but the entire set of loci enables source attribution with high probability.

4.3. Control for HhaI digestion

The tissue identification assay relies on efficient digestion of the DNA template by HhaI. Incomplete digestion might lead to erroneous results, since partially digested loci may be mistaken as being partially methylated. In this work high quality DNA samples underwent efficient digestion, as evident from testing the sample with SW14 and L98328. However, in non-pristine casework samples, digestion might not be as efficient due to the presence of various inhibitors. Therefore, for constructing a practical forensic tool it may be beneficial to develop an internal control for HhaI digestion in each reaction.

4.4. Tissue identification of aged/degraded samples

Aged and degraded samples represent an important subset of forensic casework samples. Environmental factors can damage the DNA and therefore can potentially interfere with DNA-based tissue identification. DNA methylation is a stable chemical modification that is not expected to change as a result of environmental factors commonly encountered in casework samples. However, degradation of DNA might lead to erroneous tissue identification because methylated loci that are not amplified due to degradation can be mistaken to be unmethylated. Therefore, tissue identification should employ small amplicons which are less likely to be affected by DNA degradation. We used amplicons smaller than most STR profiling amplicons, and indeed our tissue identification assays were successful on degraded samples, including a sample that showed only three typed alleles. The quality of the STR profile or a real-time assay with different-sized amplicons could be used as an indicator of the degree of degradation.

4.5. Possible advantages of DNA methylation-based tissue identification

4.5.1. Potential no consumption of evidence

Most existing tissue determination methods consume a portion of the evidentiary material, whereas a DNA-based assay does not necessarily consume physical material (if multiplexed) because it targets extracted DNA. In samples with sufficient material, a forensic scientist may use a part of the sample for tissue identification, and a different portion for profiling. However, in cases where a forensic scientist is unsure whether there is sufficient material for separate profiling and tissue identification assays, the entire sample would likely be used for STR profiling. This assay offers the possibility that the forensic scientist no longer need be faced with making a decision of which assay to perform when the sample is extremely limited. The results show that methylation loci and STR typing can be analyzed in a single reaction using commercially available STR profiling kits.

4.5.2. Integratable standardized procedures

Current tissue identification methods employ a variety of procedures, each requiring specific knowledge, reagents and equipment. In contrast, the DNA methylation-based tissue identification assay was designed such that from the forensic scientist's perspective it is very similar to DNA profiling, i.e. the same standard equipment is used, and no special training is required to carry out the protocol. In addition, DNA methylation-based tissue identification uses the same standard protocol regardless of the specific tissue(s) to be assayed (e.g. semen, blood).

4.5.3. Analysis of multiple tissues in a single test

Current protein-based methods are designed to analyze one tissue at a time and require the forensic scientist to make an assumption regarding the source of the sample. Incorrect assumptions can result in additional consumption of sample or deciding to not proceed with a second test because subsequent DNA profiling could be compromised. A DNA methylation-based tissue identification (similarly to RNA-based identification) can perform multiple tissue analyses in a single test without requiring any prior knowledge or decision making regarding the nature of the sample.

4.5.4. High sensitivity and specificity

A DNA methylation-based tissue identification assay requires a minute amount of DNA, comparable to that required for DNA profiling. In contrast protein-based methods generally require much larger amounts of material and RNA-based systems still consume additional material. The tissue identification assay was demonstrated using 1 ng of template DNA. The sensitivity of the assay is likely lower and future studies will test the range of sensitivity. This assay described herein could possibly improve the efficacy of low copy number (LCN) samples. The tissue source of most LCN samples (except possibly for example bones and teeth) is often unknown and the relevancy of the biological sample can be questioned. Although stochastic effects which are more pronounced in LCN samples might decrease the precision of tissue identification, similarly to that of STR typing, methylation might be another parameter to improve the utility of LCN typing as well as provide some indication whether the sample being analyzed is the result of contamination of amplified materials (amplified products would not show any methylation patterns [20]). If applicable to LCN samples, tissue source identification capability could help resolve or at least better explain some results of LCN typing that may be the result of secondary transfer. Furthermore, a DNA methylation-based assay can have higher specificity than protein-based assays, for example semen detection was indicated only for semen samples, and not for male urine samples, which are known to cross react with some of the commercial kits for semen detection (however, additional testing needs to be done to confirm this observation).

4.5.5. Highly applicable and robust assay

DNA tends to be more stable than protein in samples. Because the presence of intact and/or functional protein is required, it is likely that protein-based assays will fail to detect true targets well before DNA-based assays. A DNA methylation-based assay would more likely be expected to work on samples where DNA profiling is successful.

4.5.6. Numerical, user-independent results

Most protein-based commercial kits for tissue identification provide non-quantitative results, often in the form of bands that become visible upon detection of the target. Analysis in such assays is based on subjective human decision, often within a specified time frame, and can be problematic in borderline situations. Moreover, there is no statistical confidence level associated with either a positive or negative result in these assays. The fact that the DNA methylation-based tissue identification assay calculates quantitative ratios potentially allows for calculating an associated statistical confidence level of the result. In other words, the assay may allow identification of the most likely tissue source in a user-

independent manner and calculation of the likelihood that its output is correct/incorrect. This is somewhat analogous to DNA profiling, which can indicate an association between two samples (e.g. a comparison between a DNA profile from an evidence sample and a suspect reference sample) and provide significance of the association with a random match probability or likelihood ratio calculation. A similar approach for tissue identification can increase its reliability and thus credibility in legal proceedings.

4.5.7. Multiplexing with DNA profiling

The DNA methylation-based assay was designed such that it can be combined with DNA profiling. The combined assay can have several advantages over a stand-alone assay: it requires less sample manipulation and can be automated, thereby reducing labor, the risk for contamination, and human error—all of which are desirable qualities for an assay [4]. As an example a confirmatory DNA methylation-based semen detection assay can eliminate the need for manual microscopic inspection, which is labor-intensive. In addition, the fact that the tissue identification assay can indicate not just the presence or absence of a specific tissue, but also a quantitative contribution of that tissue in the sample can be important for mixture analysis by linking between a specific profile and its corresponding tissue source.

In conclusion, the methylation-based tissue identification assay described herein has several salient features which demonstrate that the assay has promise for forensic applications. It provides a single format system with high specificity that can identify several different tissues in an automatable fashion. It also can be combined with existing STR typing platforms and assays to reduce sample consumption, labor and cost of analyses. While a full validation study is still required, this proof-of-concept study suggests that tissue identification by methylation patterns will likely be a viable tool for forensic analyses of biological materials.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.fsigen.2010.12.001.

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